

Experiment-3:Temporal-Causal GraphRAG Reasoning and Next-TTP Prediction

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1. Aim:

To extract data such as Head entity, Relation, Tail entity, Time point or time interval, Source document or evidence source, Evidence sentence or paragraph, and LLM extraction confidence using a Prompt template and LLMs and creating Knowledge Graph.

2. Introduction about MITRE ATT&CK

MITRE ATT&CK® is a globally-accessible knowledge base of adversary tactics and techniques based on real-world observations. The ATT&CK knowledge base is used as a foundation for the development of specific threat models and methodologies in the private sector, in government, and in the cybersecurity product and service community.

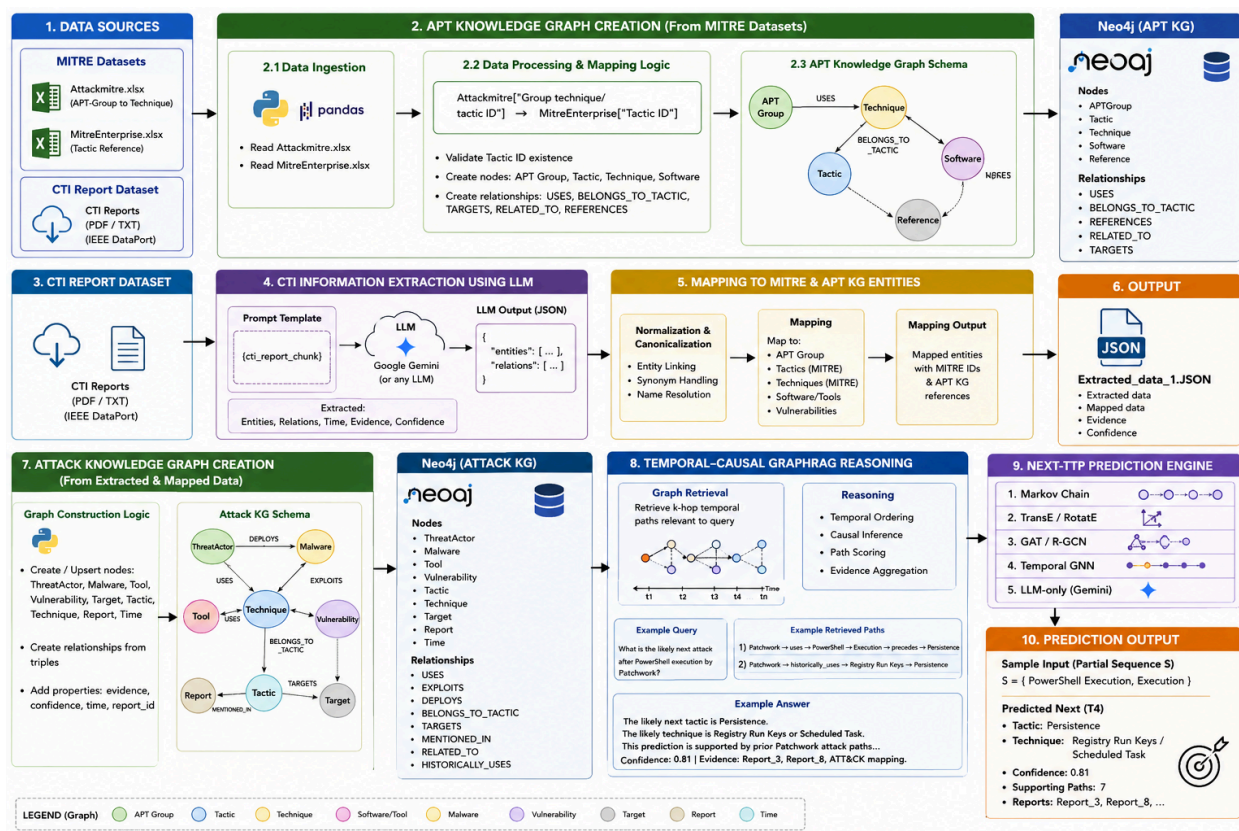
With the creation of ATT&CK, MITRE is fulfilling its mission to solve problems for a safer world – by bringing communities together to develop more effective cybersecurity. ATT&CK is open and available to any person or organisation for use at no charge.¹

MITRE ATT&CK is a widely used cybersecurity framework developed by MITRE to document and analyse adversary tactics, techniques, and procedures (TTPs) based on real-world cyberattacks. It helps organisations understand attacker behaviour, detect threats, assess risks, and improve defensive strategies. The framework organizes attacks into tactics (attacker goals), techniques (methods used), and procedures (specific implementations), enabling analysts to map and predict attack patterns. The survey paper reviews more than 50 research contributions that apply MITRE ATT&CK in areas such as behavioral analytics, red teaming, defensive gap assessment, and cyber threat intelligence enrichment. It also compares ATT&CK with other security frameworks like Cyber Kill Chain and STRIDE, highlighting its strengths in threat modeling and attack analysis across enterprise, mobile, and industrial control system environments. The paper further discusses challenges, research gaps, and future

¹ <https://attack.mitre.org/>

directions for improving cyber threat intelligence and security analytics using the ATT&CK framework.²

3. Overall workflow



4. Procedure[To do]:

1. Create an APT Knowledge graph creation
 - a. Download the dataset from the link below
 - i. <https://ieee-dataport.org/documents/mitre-attack-dataset-knowledge-graph-enhanced-rag-cyber-threat-intelligence>
 - ii. Data sets available
 1. [Attackmitre.xlsx](#) - contains columns such as
 - a. APT group ID

² Bader Al-Sada, Alireza Sadighian, and Gabriele Oligeri. 2024. MITRE ATT&CK: State of the Art and Way Forward. ACM Comput. Surv. 57, 1, Article 12 (January 2025), 37 pages. <https://doi.org/10.1145/368730> Overall workflow0

- b. Group technique/tactic ID_>referred with Tactic ID column of MitreEnterprise.Xlsx
 - c. SoftwareID
 - d. Software Techniques
 - e. Software References
- 2. [MitreEnterprise.Xlsx](#) -contains columns such as
 - a. Tactic ID
 - b. Tactic Name
 - c. Description
 - d. Mitigation Steps
 - e. Examples
- b. Write logic to create a Knowledge graph from Attackmitre.xlsx, in which the Group technique/tactic ID refers to the Tactic ID column of MitreEnterprise. Xlsx using Python
- c. The APT knowledge graph created must be stored in the Neo4j database.
- 2. Download the CTI Report dataset from the link below
 - a. <https://ieee-dataport.org//open-access/cyber-threat-intelligent-cti-dataset-generated-public-security-reports-and-malware>
- 3. Extract the data using the Below prompt template and LLM

Prompt template:

You are a cybersecurity threat intelligence analyst.

Extract CTI knowledge from the given report.

Return only valid JSON.

Extract:

1. Threat actors
2. Malware
3. Tools
4. Vulnerabilities
5. ATT&CK tactics
6. ATT&CK techniques
7. Targets
8. Temporal expressions
9. Entity-relation triples
10. Evidence sentence for each triple
11. Confidence score from 0 to 1

Input CTI text:

{cti_report_chunk}

Expected JSON output:

```
{
  "entities": [
    {
      "text": "Patchwork",
      "type": "ThreatActor",
      "canonical_name": "Patchwork APT",
      "confidence": 0.94},
  ]
}
```

```

    {"text": "PowerShell", "type": "Tool", "canonical_name": "PowerShell", "confidence":
0.91},
    {"text": "CVE-2017-11882", "type": "Vulnerability", "canonical_name": "CVE-2017-11882",
"confidence": 0.96}
],
"relations": [
  {
    "head": "Patchwork",
    "relation": "uses",
    "tail": "PowerShell",
    "time": "unknown",
    "evidence": "PowerShell is a common tool for Patchwork.",
    "confidence": 0.89
  },
  {
    "head": "Patchwork",
    "relation": "exploits",
    "tail": "CVE-2017-11882",
    "time": "unknown",
    "evidence": "Patchwork typically uses CVE-2017-11882...",
    "confidence": 0.91
  }
]
}

```

4. LLM models to use

a. LLM-only

i. Google Gemini(or any LLM model)

5. The CTI report-extracted data has to be mapped to MTREE & ATTK Framework parameters, such as APT group, TTP, etc., that are created using the APT Knowledge graph.
6. Store the extracted data and mapped data into Extracted_data_1.JSON
7. Create the attack Knowledge graph using Neo4j from the file Extracted_data_1.JSON
8. Create Temporal-Causal GraphRAG Reasoning over the created Knowledge graph(APT and Attack Knowledge graph) using the prompt question(Sample given below)

Example query:

What is the likely next attack technique after PowerShell execution by Patchwork?

Retrieved graph paths:

```

Patchwork -> uses -> PowerShell
PowerShell -> belongs_to_tactic -> Execution
Execution -> precedes -> Persistence
Patchwork -> historically_uses -> Registry Run Keys
Registry Run Keys -> belongs_to_tactic -> Persistence

```

Expected answer:

```

The likely next tactic is Persistence.
The likely technique is Registry Run Keys or Scheduled Task.

```

This prediction is supported by prior Patchwork attack paths where PowerShell execution was followed by persistence establishment.

Confidence: 0.81.

Evidence: Report_3, Report_8, ATT&CK mapping.

9. Next-TTP Prediction has to be made by using the knowledge graph created from the Attack and APT Knowledge graph using the following algorithms:

- a. Markov Chain
- b. TransE / RotatE
- c. GAT / R-GCN
- d. Temporal GNN
- e. LLM-only
 - i. Google Gemini(or any other model)

10. Example prediction to be done

Sample Input: partial attack sequence $S = \{T1, T2, T3\}$ or any attack details has to be provided and it has to provide next tactics or technique used by attacker

Sample Output: next likely ATT&CK tactic or technique T4

5. Result Tables

Model	Hits@1	Hits@3	MRR	F1
Markov Chain				
TransE / RotatE				
GAT / R-GCN				
Temporal GNN				
LLM-only				
-Google Gemini or any other model				

6. Github Link